THE LODAY REALIZATION OF THE ASSOCIAHEDRON: ITS HYPERPLANE, ITS FACETS AND ITS FACE LATTICE

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Editorial note. The author dismisses clauses (1) and (2) of the statement with 'This is straightforward from the definition.' and 'This is straightforward too.' (proof lines 697-698), and both sentences are kept verbatim inside the counted span. The remaining three clauses, including the decisive combinatorial description of the polytope, are proved inside the span; the omission concerns the definitional clauses only, so the item's core obligation stays with this card.

Editorial note. The proof opens by saying that the proposition is a weighted version of the cited results of Loday, except for clause (5) which prompted the generalisation; the citation is a bibliography entry, not a labelled item, so no navigation stub was needed.

Allowed standard prerequisites (editorial).

- The Loday realization K_omega of the associahedron and its points M(t, omega) indexed by planar trees. Set up in the paragraphs preceding the counted statement, and recalled inside the proof at lines 700-701 and 733-734. Supplies the polytope whose hyperplane, facets and face lattice the statement describes.
- The weighted versions of the classical Loday and Forcey results on realizations of the associahedron. Cited inside the counted proof at line 694. The author states that the proposition is a weighted version of those results, except for clause (5).

Proposition 1. *The Loday realization K_ω satisfies the following properties.*

(1) *It is contained in the hyperplane with equation*

$$\sum_{i=1}^{n-1} x_i = \sum_{1 \leq k < l \leq n} \omega_k \omega_l.$$

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(2) *Let $p+q+r=n$ with $2 \leq q \leq n-1$. For any $t \in \mathrm{PBT}_n$, the point $M(t,\omega)$ is contained in the half-space defined by the inequality*

$$x_{p+1} + \cdots + x_{p+q-1} \geq \sum_{p+1 \leq k < l \leq p+q} \omega_k \omega_l,$$

with equality if and only if the tree t can be decomposed as $t = u \circ_{p+1} v$, where $u \in \mathrm{PBT}_{p+1+r}$ and $v \in \mathrm{PBT}_q$.

(3) *The polytope K_ω is the intersection of the hyperplane of* (1) *and the half-spaces of* (2)*.*

(4) *The face lattice $(\mathscr{L}(K_\omega), \subset)$ is equal to the lattice $(\mathrm{PT}_n, \subset)$ of planar trees with n leaves.*

(5) *Any face of a Loday realization is isomorphic to a product of Loday realizations, via a permutation of coordinates.*

Proof. This proposition is a weighted version of the results of [13], except for Point (5), which actually prompted the introduction of this more general notion.

(1) This is straightforward from the definition.

(2) This is straightforward too.

(3) Let us denote by P the polytope defined by the intersection of the hyperplane of (1) and the half-spaces of (2). One can see that the points $M(t,\omega)$, for $t \in \mathrm{PBT}_n$, are vertices of P, since they are defined by a system of $n-1$ independent linear equations: the one of type (1) and $n-2$ of type (2). In the other way round, any vertex of P is characterized by a system $n-1$ independent linear equations with the one of type (1) and $n-2$ of type (2). We claim that any pair of equations

$$x_{p+1} + \cdots + x_{p+q-1} = \sum_{p+1 \leq k < l \leq p+q} \omega_k \omega_l$$

and

$$x_{p'+1} + \cdots + x_{p'+q'-1} = \sum_{p'+1 \leq k < l \leq p'+q'} \omega_k \omega_l$$

of type (2) appearing here is such that the corresponding intervals

$$[p+1, \ldots, p+q-1] \quad \text{and} \quad [p'+1, \ldots, p'+q'-1]$$

are either nested or disjoint. If not, we are in the configuration:

$$p+1 < p'+1 \leq p+q+1 < p'+q'+1.$$

Using these equalities and the defining inequalities of P, one can get

$$\begin{aligned} x_{p'+1} + \cdots + x_{p+q-1} &\leq \sum_{p+1 \leq k < l \leq p+q} \omega_k \omega_l + \sum_{p'+1 \leq k < l \leq p'+q'} \omega_k \omega_l - \sum_{p+1 \leq k < l \leq p'+q'} \omega_k \omega_l \\ &= \sum_{p'+1 \leq k < l \leq p+q} \omega_k \omega_l - \sum_{\substack{p+1 \leq k \leq p' \\ p+q+1 \leq l \leq p'+q'}} \omega_k \omega_l < \sum_{p'+1 \leq k < l \leq p+q} \omega_k \omega_l, \end{aligned}$$

which contradicts the definition of P. The solution of a system of equations as above, where the defining intervals are nested or disjoint, is a point $M(t,\omega)$, with $t \in \mathrm{PBT}_n$.

(4) Point (2) shows that the facets of K_ω correspond bijectively to two-vertex planar trees: the facet labeled by $c_{p+1+r} \circ_{p+1} c_q$, for $p+q+r=n$ and $2 \leq q \leq n-1$, is the convex hull of the points $M(t,\omega)$ associated to planar binary trees of the form $t = u \circ_{p+1} v$, for $u \in \mathrm{PBT}_{p+1+r}$ and $v \in \mathrm{PBT}_q$. Any face of K_ω of codimension k, for $0 \leq k \leq n-2$, is defined as an intersection of k facets. The above description of facets gives that the set of faces of codimension k is bijectively labeled by planar trees with k internal edges: the face corresponding to such a planar tree t is the convex hull of points

$M(s,\omega)$, for $s \subset t$. With the top dimensional face labeled by the corolla c_n, the statement is proved.

(5) The proof of the above point shows that it is enough to treat the case of the facets. Let $p+q+r=n$ with $2 \leq q \leq n-1$. We consider the following two weights

$$\overline{\omega} \coloneqq (\omega_1, \ldots, \omega_p, \omega_{p+1} + \cdots + \omega_{p+q}, \omega_{p+q+1}, \ldots, \omega_n) \quad \text{and} \quad \widetilde{\omega} \coloneqq (\omega_{p+1}, \ldots, \omega_{p+q}).$$

The image of $K_{\overline{\omega}} \times K_{\widetilde{\omega}} \hookrightarrow K_\omega$ under the isomorphism

$$\begin{array}{rccc} \Theta : & \mathbb{R}^{p+r} \times \mathbb{R}^{q-1} & \xrightarrow{\cong} & \mathbb{R}^{n-1} \\ & (x_1, \ldots, x_{p+r}) \times (y_1, \ldots, y_{q-1}) & \longmapsto & (x_1, \ldots, x_p, y_1, \ldots, y_{q-1}, x_{p+1}, \ldots, x_{p+r}) \end{array}$$

is equal to the facet labeled by the planar tree $c_{p+1+r} \circ_{p+1} c_q$. □

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